

Fig. 4: Creating a new account in Blynk



Fig. 6: Selecting the hardware



Fig. 8: Pressing Create button



Fig. 10: LCD widget setting



Fig. 5: Creating a new project window



Fig. 7: Typical authorisation token

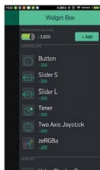


Fig. 9: Widget Box window



Fig. 11: Play button

ino sketch and upload it to Arduino Uno board. Change your IP address in IPAddress IP (your IP address) in the sketch. In this example, use your Ethernet shield and Arduino board to create a simple Web server. Using the Ethernet library, your device will be able to answer an HTTP request with the Ethernet shield.

After navigating to the Ethernet shield's IP address, Arduino will respond through HTML browser and will be ready to accept input values from analogue pins (A0 through A5) of Board1.

Step 4. Connect your mobile with Wi-Fi. Download and install Blynk app from Google Play store. Then, create a new Blynk account (Fig. 4). This account is separate from the accounts used for Blynk Forums, in case you already have one.

An account is needed to save your projects and have access to these from multiple devices anywhere in the world. It is also a security measure. We recommend using

a real email address because it will simplify things later.

Step 5. After you have successfully logged into your account, start by creating a new project (Fig. 5), and give it a name.

Step 6. Select the hardware model you intend to use. In this



Fig. 12: Play mode

case it is Arduino Uno (Fig. 6).

Step 7. Authorisation (or Auth) token is a unique identifier that is needed to connect the hardware to your smartphone. Every new project you create will have its own Auth token (Fig. 7). Click email button and the token will be sent to the email address you used for registration. Use this token in `auth[] = "your token"` in `pollution.ino` file.

Download Blynk library (Blynk_v0.3.4.zip) from <https://github.com/blynkkk/blynk-library/releases/tag/v0.3.4>, and include it in Arduino IDE. Compile and upload `pollution.ino` code to Arduino board.

Step 8. Press Create (Fig. 8).

Step 9. Your project canvas is empty at the moment. Tap anywhere on the canvas to open the widget box; all available widgets are located here. Add widgets (Fig. 9).

Add the following widgets in the settings and make the screen look like a digital dashboard as shown in Fig. 2. This includes adding the LCD (Fig. 10), LED, on/off switch, pushbuttons and RTC widgets.

Step 10. Run the project. When you are done with the settings, press Play (Fig. 11). This will switch the display from edit to play mode where you can interact with the hardware. While in play mode, you will not be able to drag or set up new widgets. Press Stop and get back to edit mode. In play mode, you will find the screen shown in Fig. 12.

EFY notes. 1. The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com.

2. Use different Wi-Fi networks for your Android mobile and Arduino sensor board for a proper IoT based test. **EFY**



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